



北京大学
PEKING UNIVERSITY

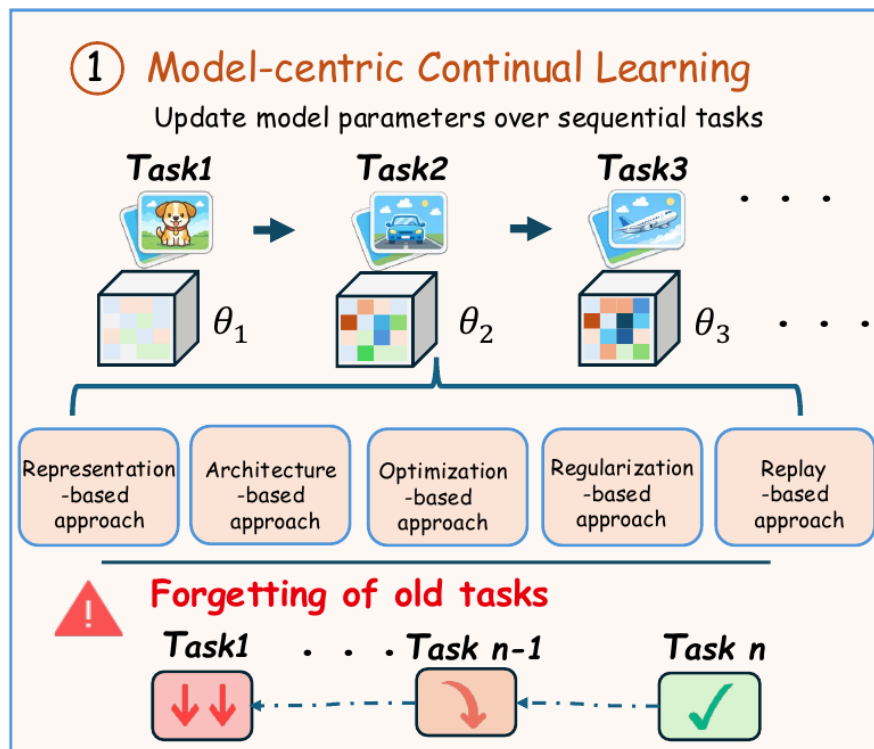
HARNESS CONTINUAL LEARNING: CONTINUAL ADAPTATION BEYOND MODEL PARAMETERS

汇报人：洪秋詠

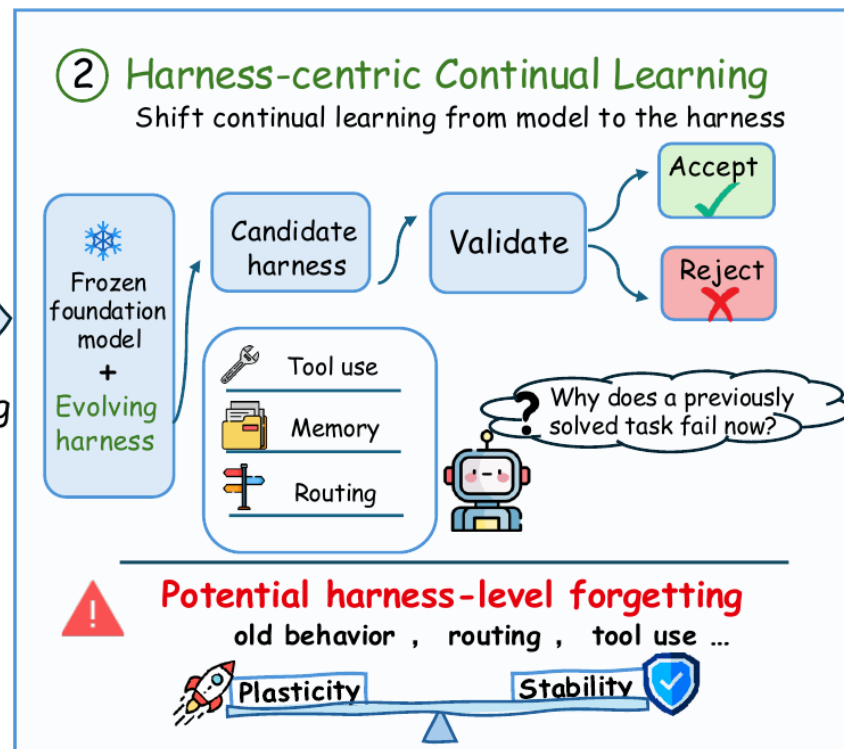
汇报日期：2026/09/02

1 模型冻结，系统仍会遗忘

Harness 漂移



Learning
object
shift



- 旧策略被覆盖
- Router发生偏移
- ...

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

1 Harness自进化范式

基本信息

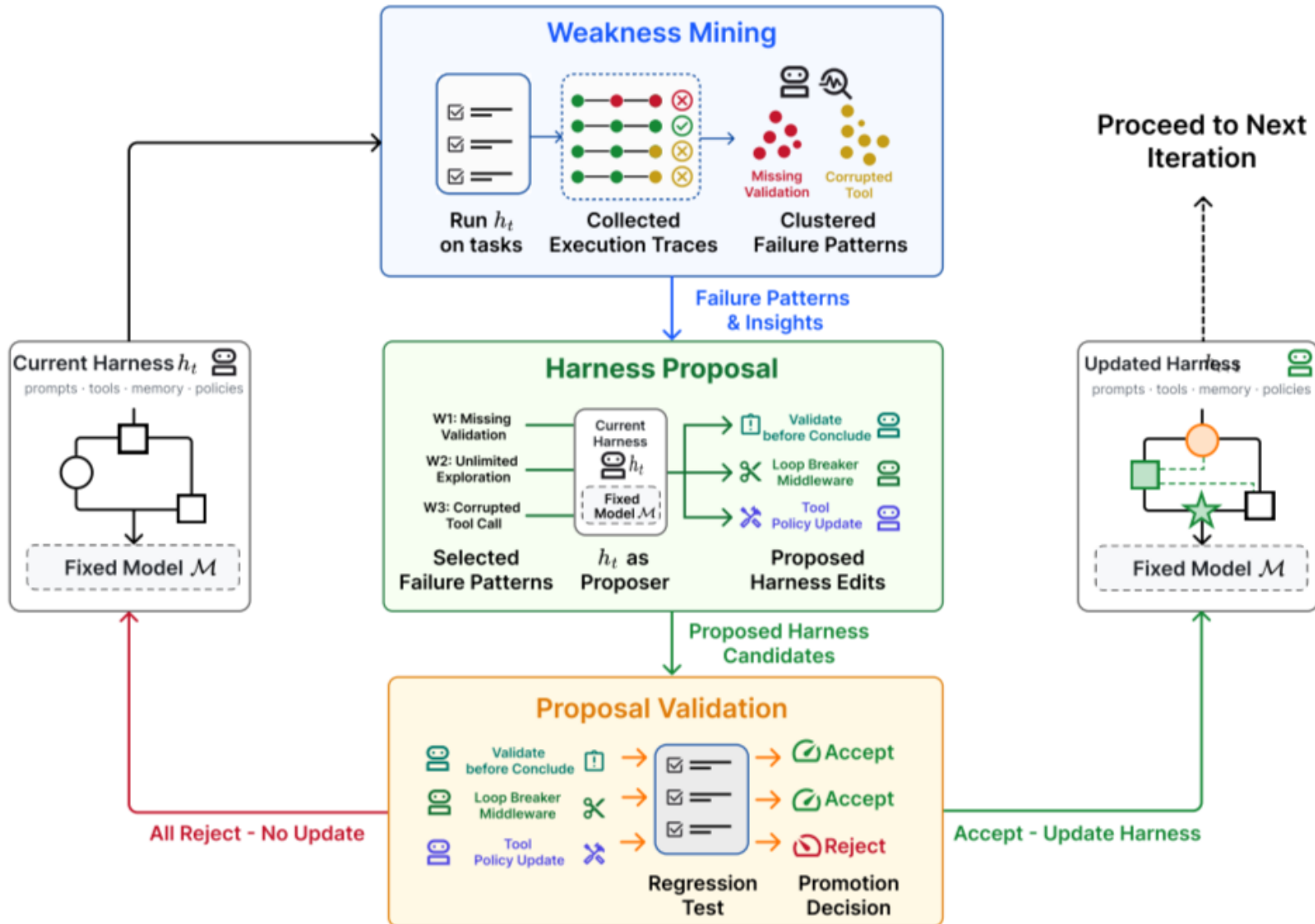
研究背景

研究思路

研究结果

结论与限制

进一步讨论



HCL: 从会不会更新到什么时候更新可以被提交

2 Harness部件拆分与形式化定义

Table 1: Execution functions and updatable contents of the four jointly versioned components in the deployed harness H_n .

Component	Function during execution	Contents updated in HCL
Task Interface I_n	Transforms raw interactions into structured representations.	Prompts, task templates, and parsing and normalization rules.
Experience Memory M_n	Provides concrete interactions and abstract guidance for reuse.	Raw interaction records and LLM-generated Abstract Memory entries.
Capability Map C_n	Provides external operations and reusable inner skills.	Inner skills extracted from Abstract Memory.
Adaptive Router R_n	Selects and organizes memory and capabilities.	Routing prompts, selection criteria, and workflow templates.

经验 e_n : input u 、task representation i 、execution context z 、outcome y 、feedback f

$$\mathbf{e}_n = (\mathbf{u}_n, \mathbf{i}_n, \mathbf{z}_n, \mathbf{y}_n, \mathbf{f}_n)$$

Optimizer $\mathcal{O}_{F_\theta}$: 用冻结 foundation model 生成 candidate $\hat{H}_{n+1}$

$$\tilde{H}_{n+1} = \mathcal{O}_{F_\theta}(H_n, \mathbf{e}_n)$$

Gate G : 校验性Gate

$$H_{n+1} = \begin{cases} \tilde{H}_{n+1}, & G_n = 1, \\ H_n, & G_n = 0. \end{cases}$$

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

3 HCL自进化workflow

基本信息

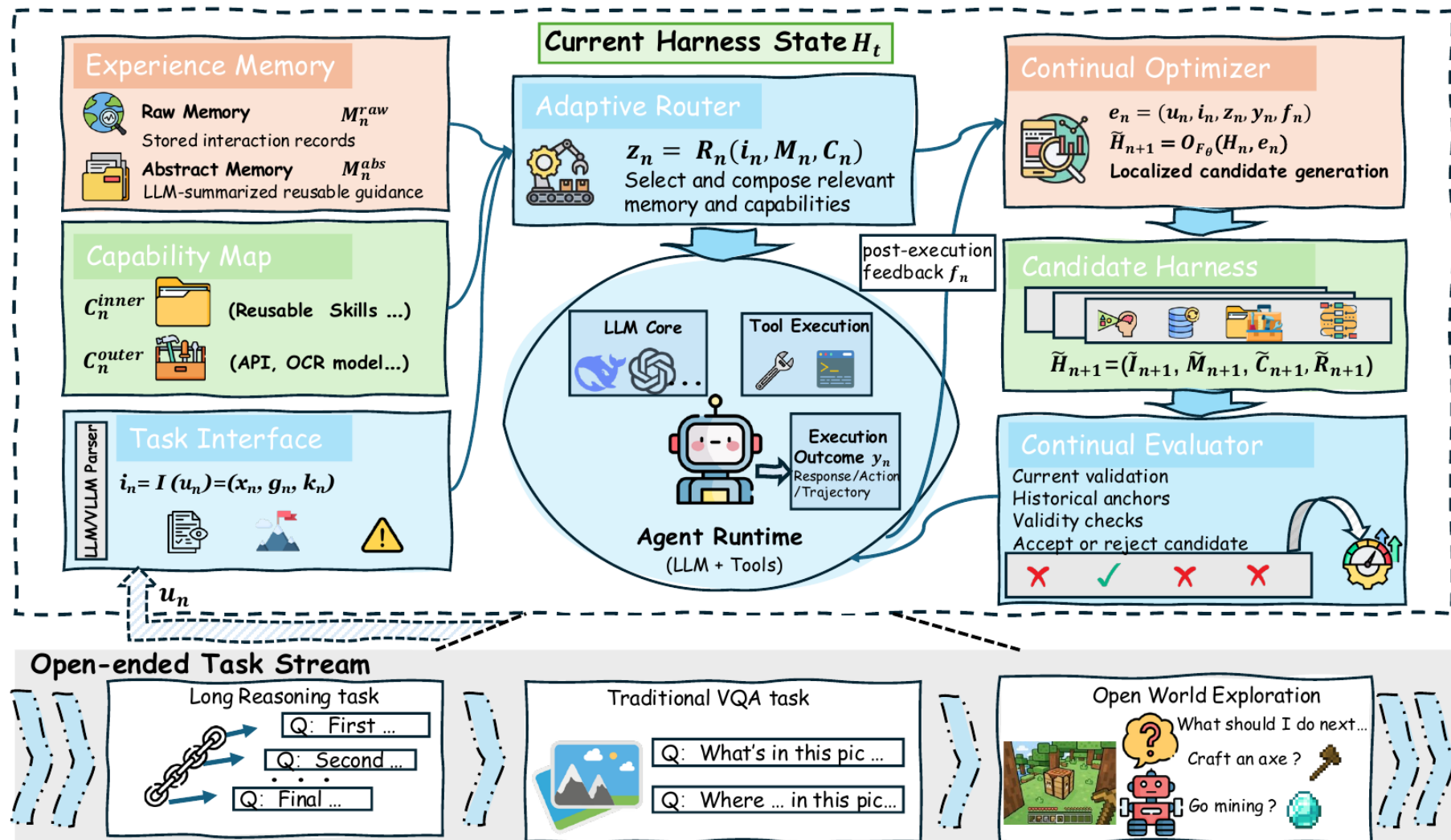
研究背景

研究思路

研究结果

结论与限制

进一步讨论



$$H_n = (I_n, M_n, C_n, R_n)$$

3 Harness Continual Learning

组件	持久内容	失败模式
Task Interface I	prompts, task templates, and parsing and normalization rules $\mathbf{i}_n = I_n(\mathbf{u}_n) = (\mathbf{x}_n, \mathbf{g}_n, \mathbf{k}_n)$	目标或约束误解
Experience Memory M	Raw Memory, Abstract Memory $M_n = (M_n^{\text{raw}}, M_n^{\text{abs}})$	检索污染 / 摘要覆盖
Capability Map C	capabilities provided by the external runtime skills acquired through continual interaction $C_n = (C_n^{\text{outer}}, C_n^{\text{inner}})$	调错工具 / skill 失效
Adaptive Router R	Routing prompts, selection criteria, and workflow templates $\mathbf{z}_n = R_n(\mathbf{i}_n, M_n, C_n)$	组合与顺序被破坏

HCL: 四层联合版本化

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

3.2 Guarded harness evolution

proposal – evaluation – commitment

CONTINUAL OPTIMIZER: CANDIDATE GENERATION

$$\tilde{H}_{n+1} = \mathcal{O}_{F_\theta}(H_n, \mathbf{e}_n)$$

- 采用简单
- 只进行选中组件的受控评估

补丁patch:

- prompts or parsing rules in the Task Interface,
- record or summarize experience in Memory,
- add or revise skills in the Capability Map,
- adjust selection and workflow rules in the Adaptive Router

CONTINUAL EVALUATOR: HISTORICAL EVALUATION AND COMMITMENT

- Current Improvement

$$\Delta_n = P(\tilde{H}_{n+1}, V_n) - P(H_n, V_n) \quad \Delta_n \geq \delta_n$$

- Historical Retention

A_n history anchor

$$D_n = \sum_{a \in A_n} \mathbf{1} \left[q(H_n, a) = 1 \wedge q(\tilde{H}_{n+1}, a) = 0 \right] \quad q(H, a) \in \{0, 1\} \quad D_n \leq B_n$$

- Validity Check

$$v_{n,\ell}(\tilde{H}_{n+1}) \in \{0, 1\}$$

Commitment Gate $G_n^{(k)} = \mathbf{1} \left[(\Delta_n^{(k)} \geq \delta_n) \wedge (D_n^{(k)} \leq B_n) \wedge \left(\forall \ell, v_{n,\ell}(\tilde{H}_{n+1}^{(k)}) = 1 \right) \right]$

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

3.3 CONNECTIONS TO MODEL-CENTRIC CL

传统 Continual Learning

学习状态: 参数 θ

更新: gradient / replay / regularization

干扰: 新梯度覆盖旧知识

评估: 任务均值与 forgetting

风险: 一次更新直接改变模型行为

Harness Continual Learning

$H_n \rightarrow$ production release

$\hat{H}_{n+1} \rightarrow$ candidate build / PR

$V_n \rightarrow$ new-feature tests

$A_n \rightarrow$ regression suite

Validity $\rightarrow$ lint / schema / sandbox

$G_n \rightarrow$ merge & deploy gate

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

4.1 Main Results

ALFWorld

Table 2: Final performance and harness-level forgetting on ALFWorld with Qwen3.5-9B as the frozen foundation model. The best and second-best results in each metric column are marked in bold and underlined, respectively.

Method	Pick	Look	Clean	Heat	Cool	Two-object	Final Avg. $\uparrow$	Avg. Fgt. $\downarrow$
Static Harness	<u>95.80</u>	66.70	25.80	26.10	9.50	58.80	47.12	—
RAG Baseline	<u>95.80</u>	83.30	41.90	<u>39.10</u>	14.30	58.80	55.56	1.74
MemP (Fang et al., 2026)	<u>95.80</u>	83.30	<u>48.40</u>	34.80	9.50	47.10	53.15	5.18
MemRL (Zhang et al., 2026c)	87.50	66.70	29.00	60.90	<u>23.80</u>	41.20	51.51	5.64
Stability-HCL (Ours)	100.00	83.30	51.60	30.40	28.60	<u>76.50</u>	<u>61.74</u>	<u>2.64</u>
Plasticity-HCL (Ours)	100.00	<u>77.80</u>	41.90	<u>39.10</u>	19.00	100.00	62.98	10.94

Minecraft Curriculum

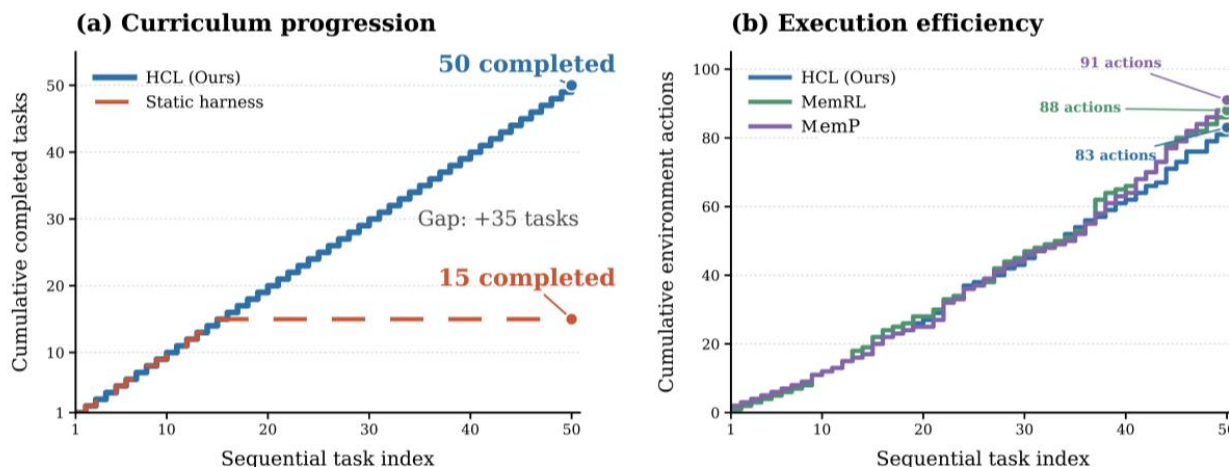


Figure 3: Curriculum progression and execution efficiency. (a) HCL completes all 50 tasks, while the Static Harness plateaus at 15. (b) Cumulative environment actions over the 50-task curriculum: HCL uses 83, versus 88 for MemRL and 91 for MemP; lower is more efficient.

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

4.1 Main Results

TEXTUAL REASONING

Table 4: Final performance after the four-task multimodal-perception stream with Qwen3.6-27B as the frozen foundation model. The Zero-shot baseline evaluates each task independently without sequential harness updates. The best and second-best results in each metric column are marked in bold and underlined, respectively.

Method	Detection	Caption	Grounding	VQAv2	Final Avg. ↑	Avg. Fgt. ↓
Qwen3.6-27B Zero-shot	4.27	25.47	43.00	84.87	39.40	–
DGG (Li et al., 2026b)	29.58	29.77	48.96	62.60	42.73	<u>0.26</u>
Plasticity-HCL (Ours)	<u>64.14</u>	<u>37.31</u>	<u>90.60</u>	<u>79.80</u>	<u>67.96</u>	0.81
Stability-HCL (Ours)	65.34	39.41	91.60	79.33	68.92	0.22

MULTIMODAL PERCEPTION

Table 4: Final performance after the four-task multimodal-perception stream with Qwen3.6-27B as the frozen foundation model. The Zero-shot baseline evaluates each task independently without sequential harness updates. The best and second-best results in each metric column are marked in bold and underlined, respectively.

Method	Detection	Caption	Grounding	VQAv2	Final Avg. ↑	Avg. Fgt. ↓
Qwen3.6-27B Zero-shot	4.27	25.47	43.00	84.87	39.40	–
DGG (Li et al., 2026b)	29.58	29.77	48.96	62.60	42.73	<u>0.26</u>
Plasticity-HCL (Ours)	<u>64.14</u>	<u>37.31</u>	<u>90.60</u>	<u>79.80</u>	<u>67.96</u>	0.81
Stability-HCL (Ours)	65.34	39.41	91.60	79.33	68.92	0.22

基本信息

研究背景

研究思路

研究结果

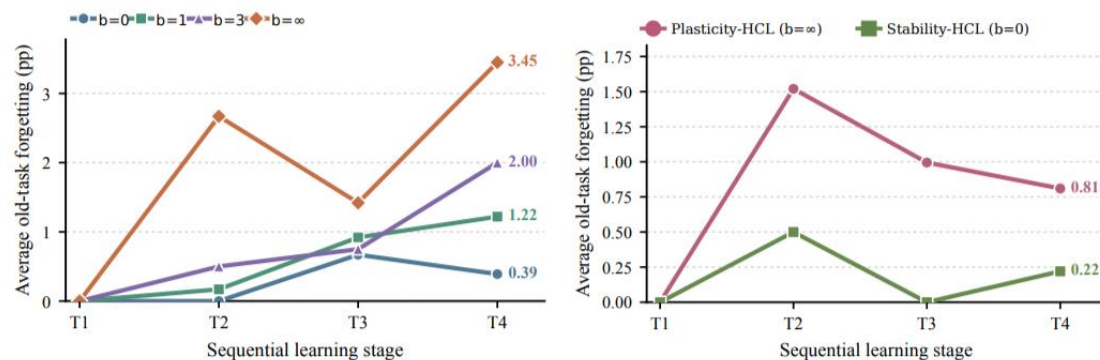
结论与限制

进一步讨论

4.2 Hyper-parameter and Ablation study

Table 5: Performance under different fixed values of b , where $B_n \equiv b$ within each run. All other experimental conditions are held constant. The best and second-best results in each metric column are marked in bold and underlined, respectively.

Historical-loss tolerance b	MuSiQue	ProofWriter	GSM8K	HotpotQA	Final Avg. $\uparrow$	Avg. Fgt. $\downarrow$
$b = 0$	<u>27.83</u>	73.33	<u>84.33</u>	59.50	61.25	0.39
$b = 1$	24.83	<u>77.50</u>	92.33	<u>59.17</u>	63.46	<u>1.22</u>
$b = 3$	26.83	79.83	83.00	58.50	<u>62.04</u>	2.00
$b = \infty$	28.33	71.00	82.00	<u>59.17</u>	60.13	3.45



(a) Textual reasoning under different fixed values of b . (b) Multimodal perception under $b = 0$ and $b = \infty$.

Figure 4: Stage-wise forgetting under different fixed historical-loss tolerances.

Table 6: Component ablation on the controlled multimodal stream. I , M , C , and R denote the Task Interface, Experience Memory, Capability Map, and Adaptive Router. A check mark indicates that the component is updated, while a cross indicates that its update is disabled. The best and second-best results in each metric column are marked in bold and underlined, respectively.

Component	I	M	C	R	Final Avg. $\uparrow$	Avg. Fgt. $\downarrow$
Zero-shot	—	—	—	—	34.84	—
w/o Interface update	×	✓	✓	✓	62.37	<u>0.11</u>
w/o Memory update	✓	×	✓	✓	62.28	0.83
w/o Capability update	✓	✓	×	✓	<u>63.12</u>	0.06
w/o Router update	✓	✓	✓	×	62.77	0.14
Full HCL	✓	✓	✓	✓	63.41	0.45

- b 越大, forgetting 基本单调上升; 但 final average 不单调。
- $b=1$ 得到最佳 63.46; $b=\infty$ 反而只有 60.13。
- 一次 commit 会改变未来检索、skill 与 feedback; 早期覆盖会破坏后续学习基座。
- retention gate 也在选择长期更优的演化路径。

- 较低 forgetting 也可能只是“改得更少”, 并非更会保留。
- 组件不正交: 禁用 Memory 同时切断 inner skill 蒸馏路径; 不同 variant 也不共享候选轨迹。

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

5 ‘零遗忘’ 保护的是有限行为契约

Conclusion

- 提出harness层面的continual learning框架，在架构层面也存在序列任务性能上的stability-plasticity bottleneck
- 提出一个在submission gate上进行约束的算法改进，并探讨了由anchor通过阈值隐性控制的平衡效果
- 通过实验揭示了harness算法对整体持续学习性能的较大影响

Limitation

- 算法较简单，与自进化范式相同
- 并没有讨论不同组件的selection比例以及各自对stability/plasticity的贡献
- Stability-HCL 仍低于 zero-shot: MuSiQue $27.6 < 35$; VQAv2 $79.33 < 84.87$
- 评测成本高，随anchors大小与candidates数量线性增长

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

6 进一步讨论

基本信息

研究背景

研究思路

研究结果

结论与限制

进一步讨论

- 本文偏为概念性，定义了Harness Continual Learning
- 相比于model continual learning在训练数据/参数更新约束/动态架构上进行方法改进，harness continual learning可以在batch building, evolver manager, **evaluator & submission(anchor/threshold/...)**进行改进
- 本文并没有对不同组件更新/以及与模型参数更新动力学&性质进行深入实验与讨论
- 目前的harness自进化比较简单，并没有进行分任务约束memory/router/schema等
- 这篇HCL的定义偏清晰的任务定义边界，或许open-domain场景的CL更符合agent的实际 workflow
- 这篇的evolving batch均为1，容易造成漂移，在改进batch大小后或许能直接减缓遗忘
- I,M,C,R被顺序更新，但组件间存在强耦合；消融实验中不同方法沿不同演化轨迹前进，难以得到严格因果归因